

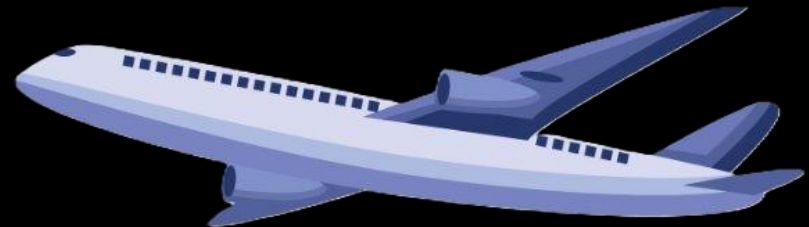
Computer Science Capstone

GULFSTREAM: AI 3D MODEL GENERATION

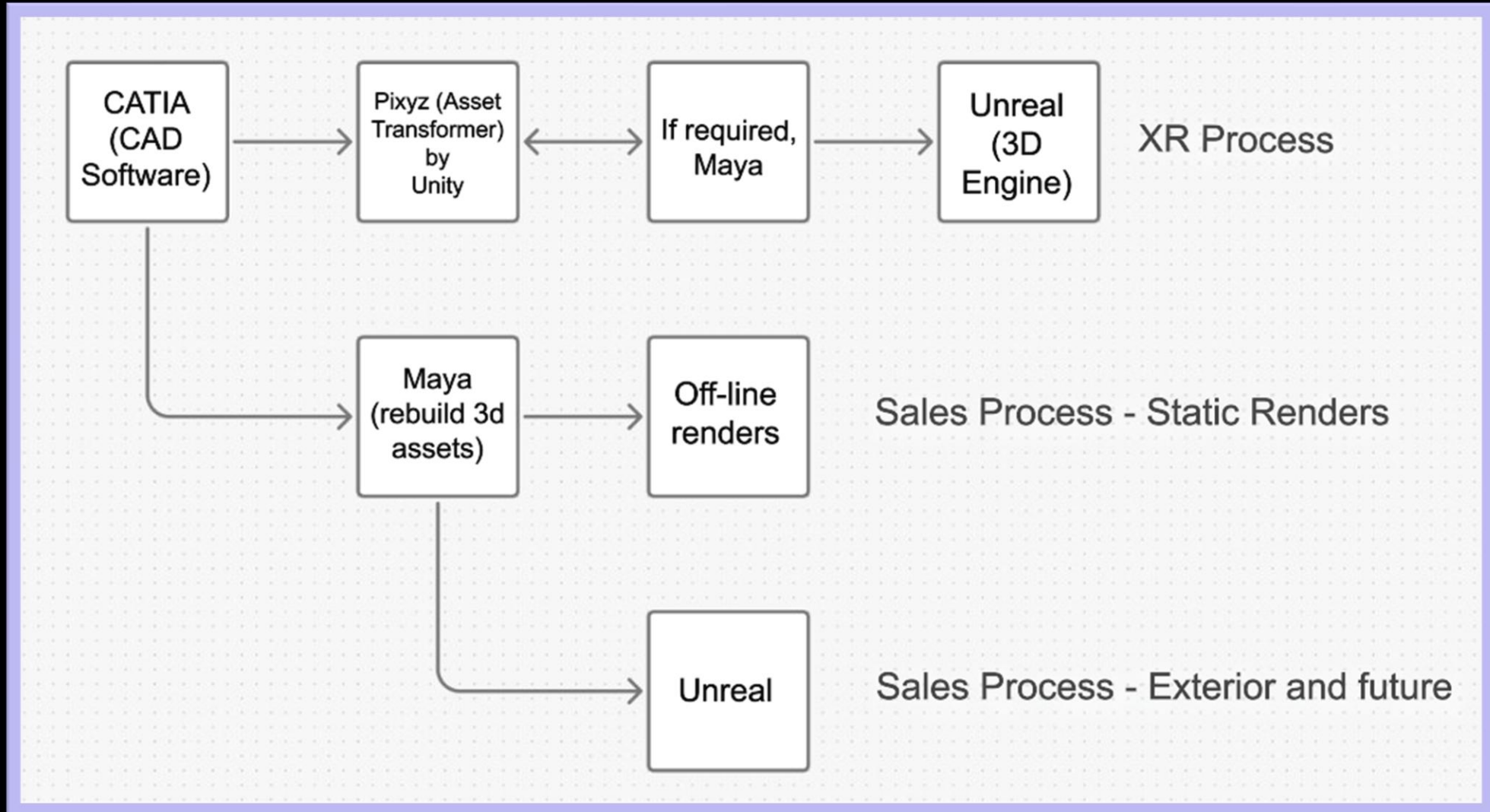
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CONTEXT

- ❑ Gulfstream is widely regarded as a leader in the business aviation industry. The company is very well known for selling business (private) jets to clients.
- ❑ Their product lineup consists almost entirely of luxury business jets (e.g., G400, G700), which are specifically built for private travel and customized interiors.
- ❑ Currently, Gulfstream's designers rely on traditional 3D modeling workflows when performing interior design tailored to client specifications.



GULFSTREAM'S WORKFLOW



THE PROBLEM

Time	Labor	Service
• Traditional 3D modeling and review cycles are slow to scale. • Schedule deadline or time constraints is one of the most cited workflow barriers. (Autodesk, 2025) • Simple changes can still require repeated remodel, re-export, and scene validation.	• Skilled design labor is difficult to scale. • 61% say technical talent is hard to find; 58% say talent shortages already limit growth. (Autodesk, 2025)	• Slow iteration delays client-facing design comparisons and approvals.

THE SOLUTION

- ❑ According to Render Edge Studio, AI-enhanced workflows can reduce rough draft render times from 4 –12 hours to as little as 15 –90 seconds —representing a 95 –99% reduction in rendering time.
- ❑ 86% architects/designers that use AI-enhanced workflows report time savings; medium-size firms report 10-15+ hours saved per week. (Chaos, 2026)
- ❑ Our application turns that into a browser-to-Unreal pipeline: prompt or image input -> 3D generation -> Blender normalization -> Mesh Inspector -> Unreal export.



KEY FEATURES

DUAL 3D GENERATION SERVICES

Local

Hardware-Dependent

- Trellis (Text/Image → 3D Model)
- Z-images enhancement pipeline (sub-step): Enhances important visual details in generated images before final 3D reconstruction, helping create smoother surfaces, cleaner object shapes, and more stable 3D models.
- TripoSR (Image → 3D Model)

Cloud (New)

Hardware-Independent

- ModelsLab(Text/Image to 3D Model): Managed API-based generation used for hardware-independent text and image workflows.
- Cloud compute handles GPU-heavy generation externally, reducing local hardware dependence during evaluation and demo workflows.

AI-ENHANCED OBJECT DESCRIPTIONS

- Users can type a description of the 3D asset they want to generate.
- (New): An OpenAI-powered agent can transform text prompts into structured, technically detailed 3D asset specification, fine-tuned for realism.

INPUT

DESCRIBE THE 3D MODEL:

E.g. "A realistic chair with wooden legs and a cushioned seat"

Redo

Enhance Description



IMAGE UPLOAD & REFERENCE INPUT

- Users can upload a reference image of the 3D asset they want to generate.
- (Updated): The application supports direct reference-image uploads for image-to-3D generation through the available local or cloud model.

INPUT

GENERATE IMAGE FROM TEXT:

E.g. 'A futuristic helmet, centered, studio lighting, white background'

Generate Image

UPLOAD IMAGE:

OR



Click to upload or drag and drop

PNG, JPG, JPEG up to 10MB

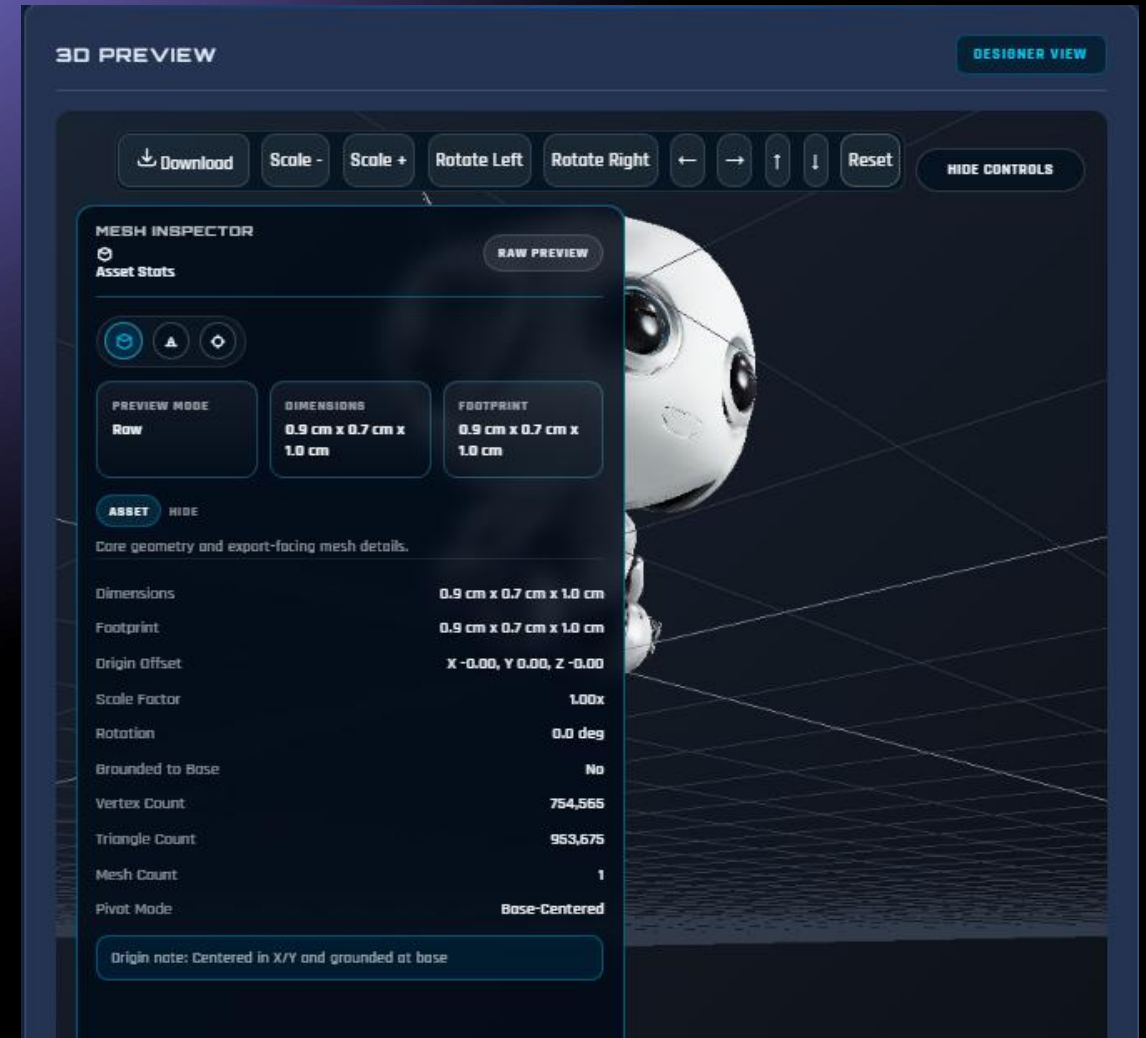
AUTOMATED NORMALIZATION (NEW)

- Designers often run into asset issues when a high-quality mesh is used across different technologies & in turn, across different formats (.FBX, .GLB, etc).
- Generated meshes are automatically processed using Blenderscripts, where the respective transformation are applied to optimize on Unreal:
 - Base-centered pivot & placement
 - Scaling
 - Rotation
 - Z-axis (up)

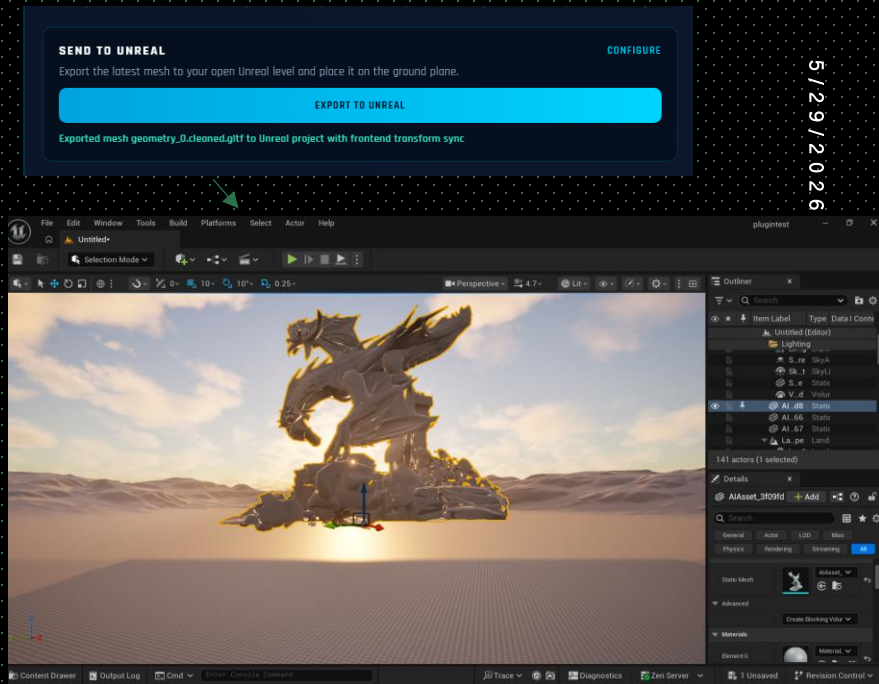
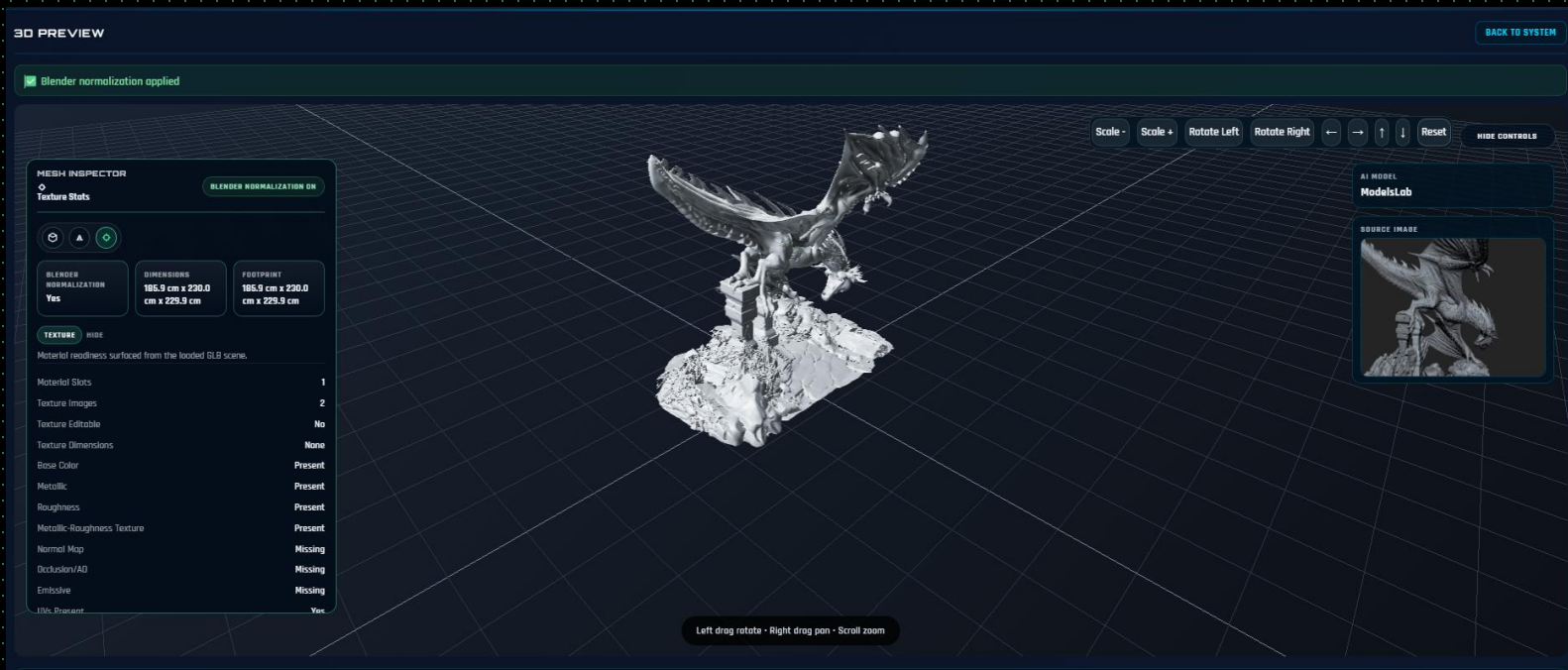
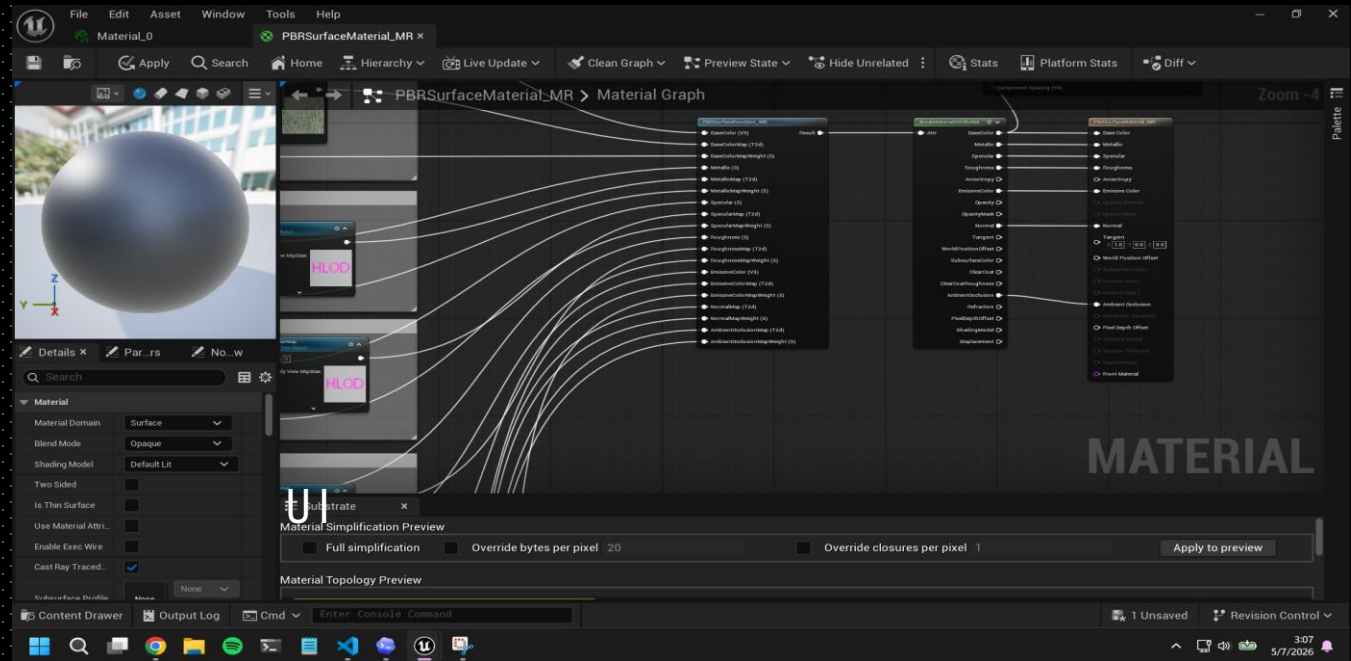
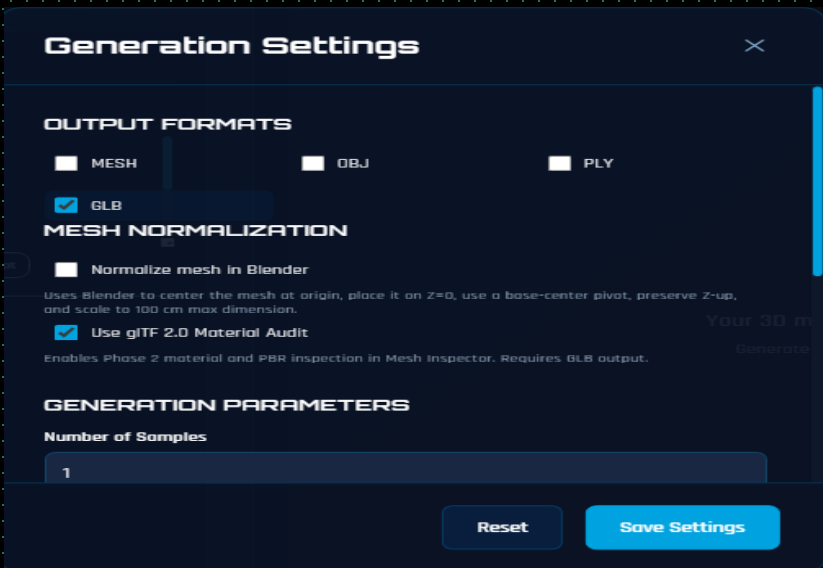


INTERACTIVE POST-PROCESSING

- Mesh Inspector: A real-time 3D visualization environment with full camera controls (orbit, zoom, pan) and transform tools (scale, rotate, translate) for validating generated models. It exposes asset statistics (dimensions, vertices, etc), normalization status, and texture/material readiness.
- Texture-aware glTF 2.0 export: Blender preserves compatible textures and material assets through the cleaned pipeline when supported.
 - This keeps the browser preview and Unreal import visually consistent whenever the target format supports it.



TEXTURE MAPPING



UNREAL ENGINE EXPORT

- Users can export generated 3D meshes from the browser into an active Unreal Engine session with one click.
- Handles automated mesh transfer, enabling:
 - Automated import, folder organization, and actor placement
 - Viewer transforms carry over as Unreal actor transforms
 - Faster iteration between generation, inspection, and scene placement

Unreal Integration

Link your running Unreal Editor instance. Enable the Web Remote Control plugin and keep the editor open on the same machine.

REMOTE CONTROL URL

`http://localhost:30010`

.UPROJECT PATH

`C:/Projects/MyProject/MyProject.uproject`

DESTINATION CONTENT PATH

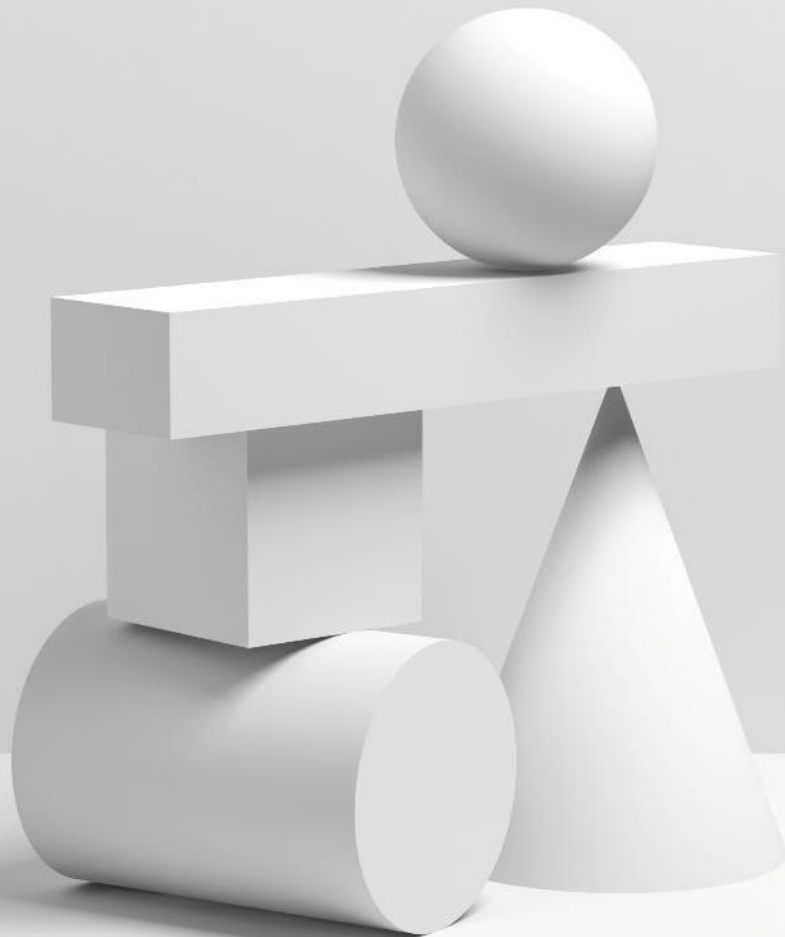
`/Game/GeneratedMeshes`

ACTOR PREFIX

`AAsset`

☒ SNAP ACTOR TO GROUND LEVEL

Test Connection Save Settings



CITATION

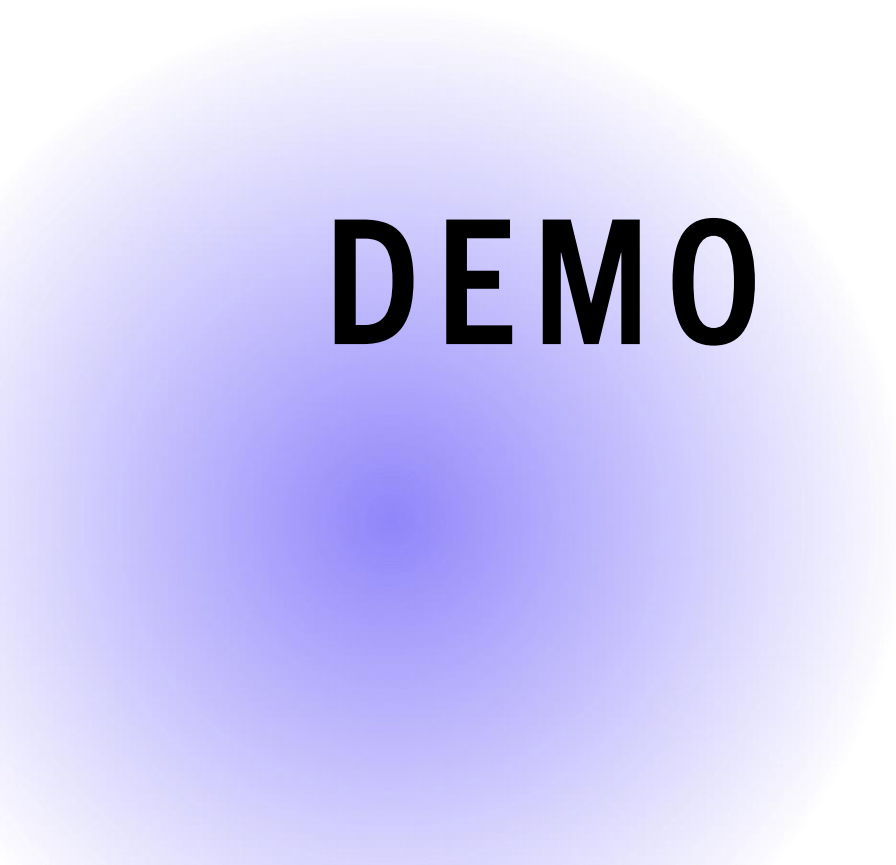
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2025. <https://www.autodesk.com/design-make/research/state-of-design-and-make-2025/digital-transformation>

Chaos. “How much time are architects really saving with AI?” April 27, 2026. <https://blog.chaos.com/how-much-time-are-architects-saving-with-ai>



DEMO